SE2205: Software Quality Assurance

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Week 01:
Basic Concepts of Software Quality Assurance

Learning Outcomes

- LO1: Distinguish the importance of software quality assurance in all the testing stages of

software development life cycle

- LO2: Learn information and requirements extraction from stakeholders, project documentation and other relevant sources
- LO3: Select appropriate testing types and techniques focus on the projects and the expected outcome
- LO4: Identify, assign priorities, plan, create and execute test cases plus measure and report the progress of testing
- LO5: Successfully identify, record and manage defects
- LO6: Gain knowledge of the importance of automated testing

Examination Criteria

Midterm Examination - 20%

Assessment - 20%

Semester End Examination - 60%

Aims of The Session

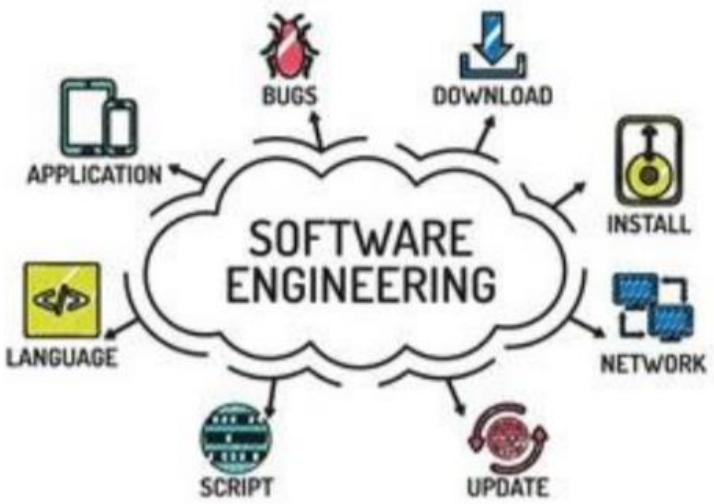
- Learn the importance of SQA
- The role of SQA in System Development Life Cycle
- Learn about software quality factors
- Study about Cost of Quality

Background

What is Software?

“Computer programs, procedures, and possibly associated documentation and data pertaining to the operation of a computer system.”

- IEEE definition-



- **Computer programs** - activate the computer to perform the required applications.
- **Procedures** - define the order and schedule in which the programs are performed, the method employed, and the person responsible
- **Documentation** - needed for developers, users and maintenance personnel.

- Development documentation

- The user's documentation

- The maintenance documentation

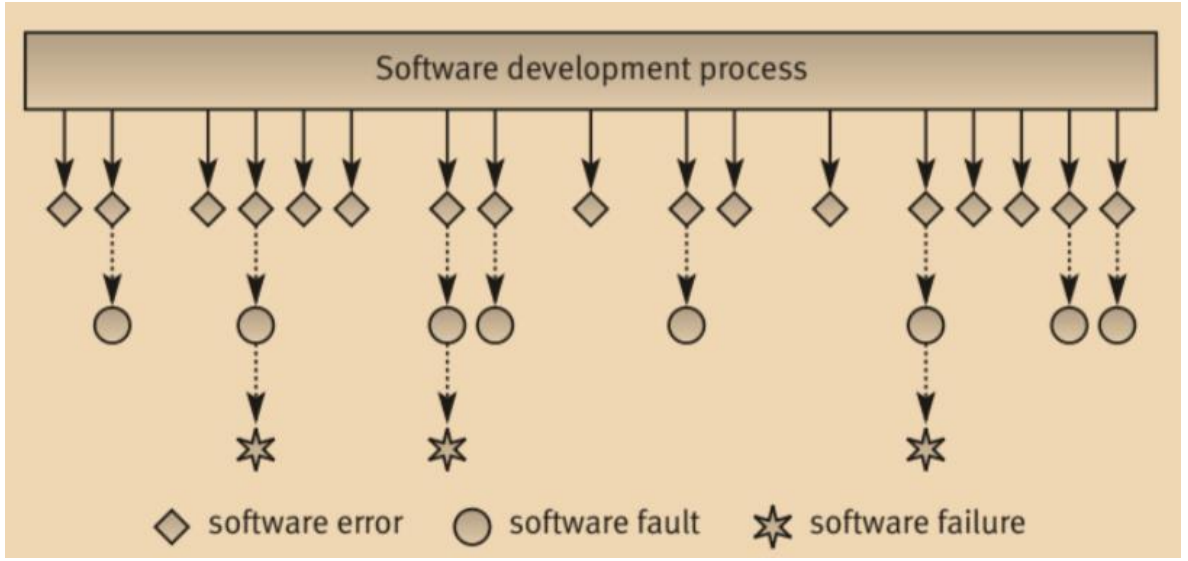
- **Data** – to make necessary changes or to keep track of the changes done

- parameters, codes and name lists that adapt the software to the needs of the specific user

- Test data

Software Faults, Error And Failures

- Origin of software **failure** lies in a software **error** made by a programmer in one or more of code lines.
- It could be **grammatical** or **logical** error
- Some erroneous code lines (software errors) affect the functionality of the software during a specific application.
- A software fault becomes a software failure only when the user tries to use the faulty application.



Classification of the Cause of Software Error

Faulty Requirements Definition
Client – developer communication failure
Deliberate deviations from software requirements
Logical Design Errors
Coding Errors
Non-compliance with documentation and coding
Shortcomings of the testing process
Procedure Errors
Documentation Errors

Software Quality

Software Quality is:

“The degree to which a system, component, or process meets specified requirements.”

Or

“ The degree to which a system, component, or process meets customer or user needs or expectations.”

- IEEE definition

Software Quality Assurance

Software Quality Assurance is:

“A planned and systematic pattern of all actions necessary to provide adequate confidence that an item or product conforms to established technical requirements.

or

A set of activities designed to evaluate the process by which the products are developed or manufactured. Contrast with quality control”

-IEEE definition

Software Quality

- Software quality assurance (SQA) is a crucial process that must be implemented during the software development life cycle.

•The primary purpose of QA is to ensure that the software meets **planned functionality, bug-free**, and it will be working **flawlessly under different circumstances.**

•QA specialists create and implement various **strategies** for software quality improvement.

•QA apply the required **testing types** to ensure that the software will function correctly.



Quality Control

Quality Control is:

“A set of activities designed to evaluate the quality of developed or manufactured product”

- IEEE definition

QA Vs QC Vs Testing

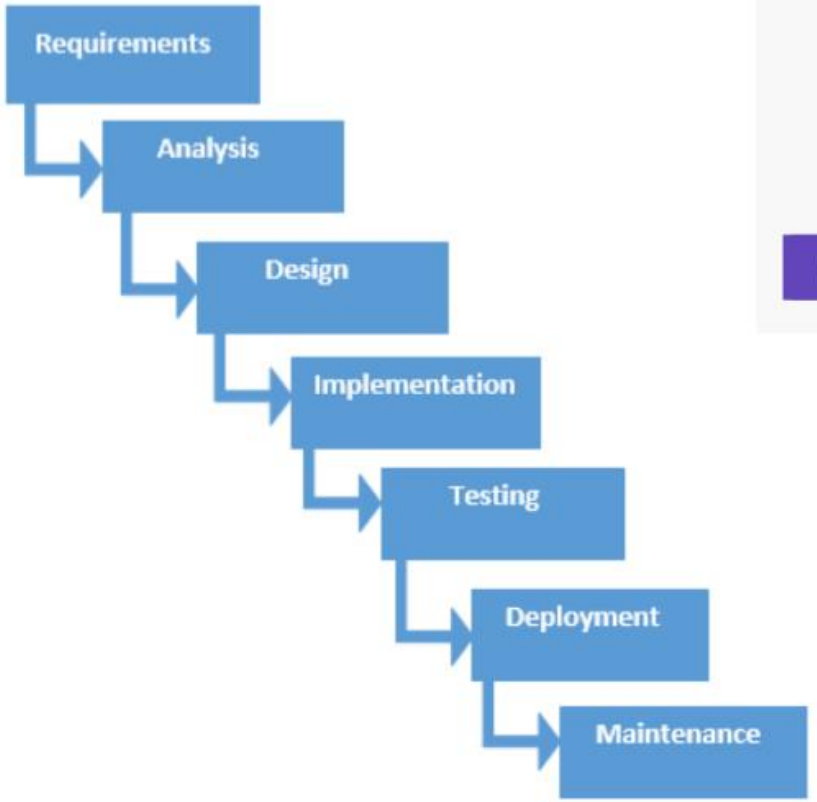




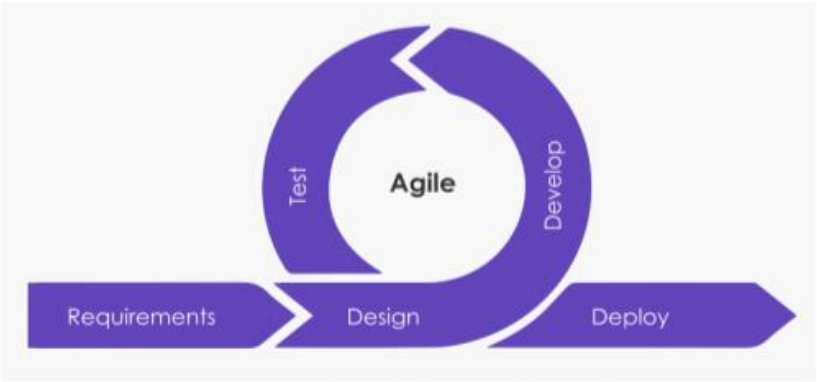
QA Vs QC Vs Testing

QA	QC	Testing
Includes activities that ensure **implementation** of processes in context to verification of developed software and intended requirements	Includes activities that ensure the verification of developed software with respect to documented requirements	Activates that ensure the identification of bugs/error/defect in the software
Focus on processes and procedures rather than conducting testing on the system	Focus on actual testing to identify bugs/defects	Focus on actual testing
Preventive activities	Corrective activities	Corrective activities
Subset of STLC (Software Testing Life Cylce)	Subset of QA	Subset of QC

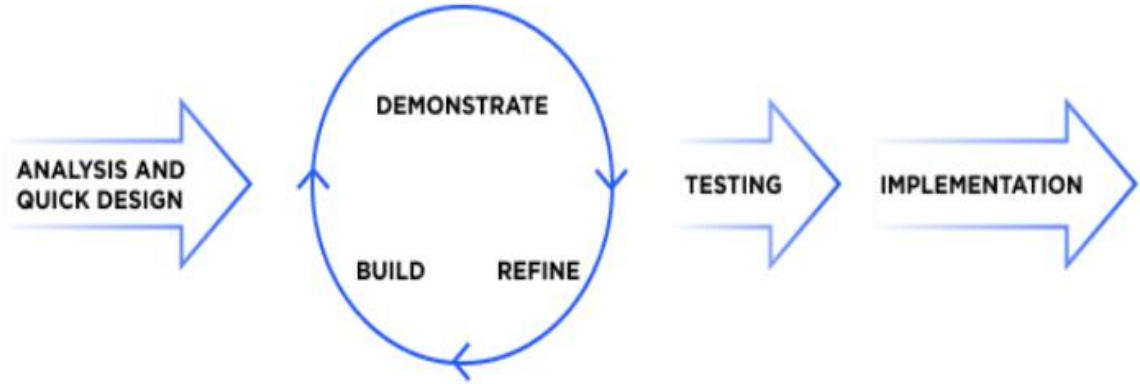
SDLC



Waterfall Methodology



Agile Methodology



RAD Methodology

Role of QA in SDLC

- **Planning**

- QA testers think like the user and speak up for a good user experience (UX).
- If a new feature is being discussed, QA can spot possible problems early, like confusion in how the feature might work.
- QA can start planning test cases

- **Design**

- QA can identify aspects of the design that might cause problems
- QA helps designers catch issues early, so developers don't have to fix them later.

- **Implementation**

- QA continues to think from the user's point of view and checks how the new feature will feel to use.

Role of QA in SDLC

- **Testing**

- QA will focus on writing test cases and bug reports

- Keeps track of all updates that comes out

- They make sure the app works well on all the devices and browsers your clients use.

- **Deployment**

- QA is on standby during deployment in case anything breaks.

- They do quick checks (called smoke testing) right after the release to make sure everything still works.

- **Maintenance**

- After deployment, bugs that slips through can be fixed

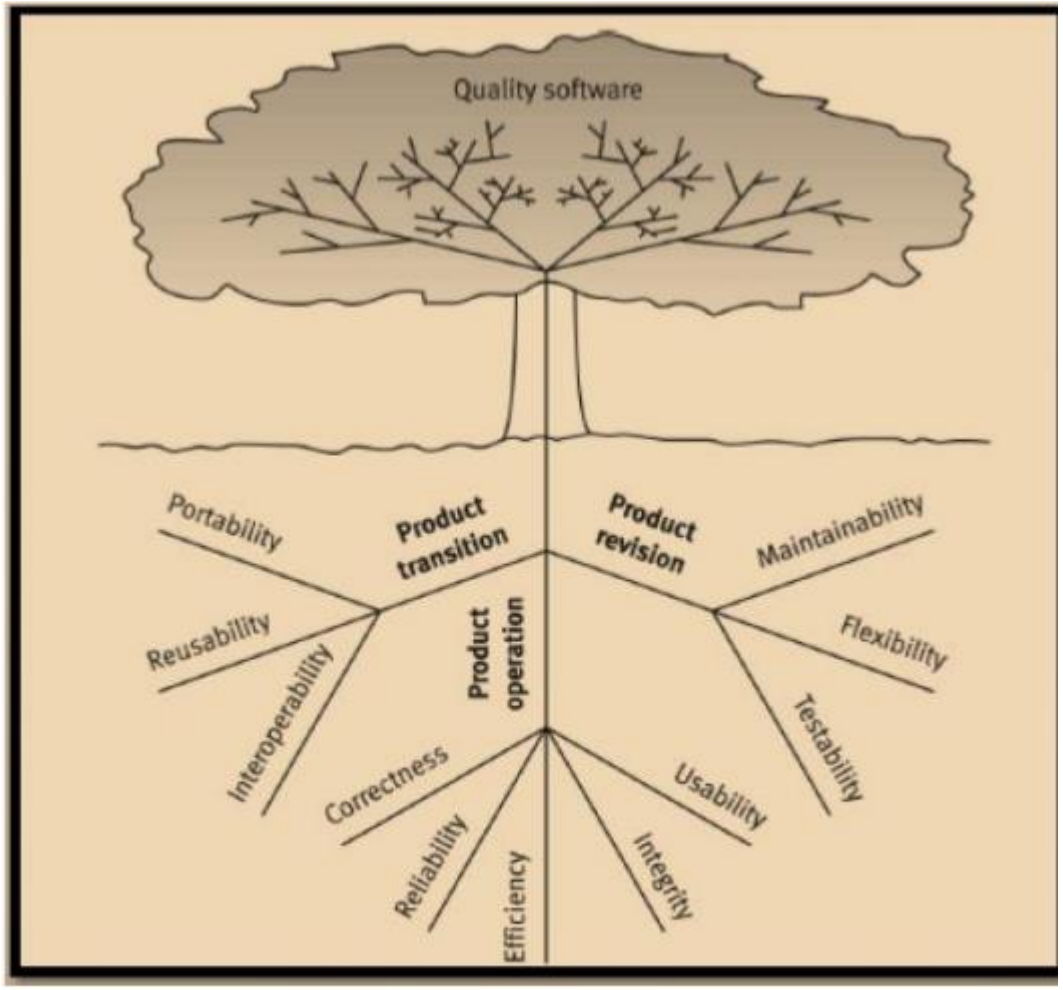
- QA also gets involved when new features or upgrades are added, to make sure they work well with the existing app.

Software Quality Factors

- **Product Operation** – factors that relate to the requirements which directly affect the daily operation of the software.

•**Product revision** – deals with minor adjustments

•**Product transition** - deals with the adaptation of to other environments and its interaction with other software systems



McCall's Factor Model Tree

Product Operations

- Correctness: Output specifications
  - The output mission
  - The required accuracy

- Reliability: Deals with failures to provide service.
  - Determine the maximum allowed SW system failure rate

- Efficiency: Deals with the resources needed to perform all the functions of the SW system in conformance to all other requirements.

- Integrity: Deals with the SW system security by preventing access to unauthorized persons.

- Usability: Deals with the scope of staff resources needed to train a new employee and to operate the SW system.

Product Revision

- Maintainability: determine the efforts in identifying the reasons for SW failures, to correct the failure, and to verify the success of the corrections.
- Flexibility: efforts required to support adaptive maintenance activities.

- Support perfective maintenance activities, such as changes and additions to the SW and adapt it to changes in the firm's technical or commercial environment.

- Testability:-Deal with the testing of the system and its operation.

- Providing predefined intermediate results and log files

- Automatic diagnostics performed to find out the performance of all components of SW system.

- Obtain a report about detected faults

Product Transition

- Portability:- adaptation of a SW system to other environments consisting

- Different HW

- Different OS

- Reusability:- Deals with using previous SW components on to new projects that are being developed.

- Save resources, shorten the project period, and provide higher quality modules.

- Interoperability:- Focus on creating interfaces with other SW systems or with other equipment firmware.

Cost Of Quality

•COQ is a method that allows an organization to determine the extent to which its resources are used for activities that prevent poor quality, that appraise the quality of the organization's products or services, and that result from internal and external failures.

•This allows an organization to determine the potential savings to be gained by implementing process improvements.



Summary

- Software Errors are caused due to logical or grammatical errors, and they may cause failures in the functionality of the software
- Quality Assurance is involved in all steps in the software development life cycle
- McCall's factors model tree provide factors that decides the quality of a software.
- The extend to which the quality of the product should be maintained is decided through Cost of Quality curve in a specific company.

Brief Of Next Lecture

- Reasons for learning software testing
- Role of a software tester
- Scope of testing and constraints
- Life cycle of Testing
- Levels of testing

Thank You